

Aditya Amol Bhongade

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📄 Profile | 📄 adityabhongade | 📄 Scholar
Nagpur, Maharashtra, India

EXPERIENCE

• IIT Dhanbad

Feb 2026 - Present

Research Intern

[Certificate]

- Under the supervision of **Dr. Partha Pratim Roy** (Dept. of CSE).
- **Field of Work:** Vision Language Models for Complex Document Understanding.
- **Skills:** Deep Learning, Large Language Models (LLMs), Vision Language Models,
- **Technologies:** PyTorch, Hugging Face, HF Transformers and Diffusers, TRL, vLLM, llama.cpp.

EDUCATION

• Yeshwantrao Chavan College of Engineering

Nov 2022 - Jun 2026

Bachelor of Technology (B. Tech) in Electronics & Telecommunication Engineering

Nagpur, India

- CGPA: 8.8/10
- **Bachelor's Project:** "Deep Learning-based Liver Fibrosis Grading System with Conversational Support through Medical LLM"

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION

- [J.1] Yogita, Dubey, Aditya Bhongade, et al. (2025). **LivSCP: Improving Liver Fibrosis Classification Through Supervised Contrastive Pretraining**. *Diagnostics*, Vol. 15, No. 24, Article 3226. DOI: 10.3390/diagnostics15243226.
- [J.2] Aditya Bhongade, et al. (2025). **Robust and Explainable Ensemble Based Framework for Liver Disease Classification Using Data Balancing and Upsampling**. *SSRG International Journal of Electronics and Communication Engineering*, Vol. 12, Issue 2, pp. 1–11. DOI: 10.14445/23488549/IJECE-V12I2P101.
- [J.3] Yogita Dubey, Aditya Bhongade, et al. (2024). **Efficient Explainable Models for Alzheimer's Disease Classification with Feature Selection and Data Balancing Approach Using Ensemble Learning**. *Diagnostics*, Vol. 14, No. 24, Article 2770. DOI: 10.3390/diagnostics14242770.
- [C.1] Aditya Bhongade, et al. (2025). **Transformer and CNN-Based Lightweight Ensemble for Retinal Disease Classification Using OCT Imaging**. In *TENCON 2025 - 2025 IEEE Region 10 Conference (TENCON)*, pp. 1068–1072. DOI: 10.1109/TENCON66050.2025.11375147.
- [C.2] Aditya Bhongade, et al. (2025). **LightSeg: A Lightweight Transformer Based Network for Lung Segmentation in Chest X-Rays**. In *2025 IEEE 6th India Council International Subsections Conference (INDISCON)*, pp. 1–6. DOI: 10.1109/INDISCON66021.2025.11253558.
- [C.3] Aditya Bhongade, et al. (2025). **Explainable Vision Ensemble for Alzheimer's Disease Classification Using Brain MRI**. In *2025 IEEE 6th India Council International Subsections Conference (INDISCON)*, pp. 1–6. DOI: 10.1109/INDISCON66021.2025.11251953.
- [C.4] Aditya Bhongade, et al. (2025). **An Efficient and Explainable Model for Heart Disease Classification Using Boosting Algorithms**. In *2025 6th International Conference for Emerging Technology (INCET)*, pp. 1–6. DOI: 10.1109/INCET64471.2025.11140243.
- [C.5] Aditya Bhongade, et al. (2024). **Retinal Blood Vessel Segmentation Using Twin U-Net Based Deep Learning Architecture**. In *2024 IEEE 21st India Council International Conference (INDICON)*, pp. 01–06. DOI: 10.1109/INDICON63790.2024.10958322.
- [C.6] Aditya Bhongade, et al. (2024). **Explainable AI Model for Medical Insurance Premium Prediction Using Ensemble Learning with SHAP Analysis**. In *2024 IEEE Pune Section International Conference (PuneCon)*, pp. 1–6. DOI: 10.1109/PuneCon63413.2024.10895118.

FUNDED PROJECTS

• SAR Image Colorization using Generative Adversarial Networks

Dec 2025 - Present

- Received funding of INR 45,000.
- Designed novel generator and discriminator neural networks for SAR image colorization.

• Deep Learning aided Face Recognition-based Authorization System

August 2025

- Received funding of INR 20,000.
- Developed a hardware-based authorization system using Raspberry Pi 4 and convolutional neural network (CNN).

HONORS AND AWARDS

• IndiaAI Fellow <i>Government of India</i> ◦ Awarded with the prestigious IndiaAI Fellowship worth INR 1 Lakh. ◦ Project Title: "Energy based Face Matching Model for ID Card Verification" • 1st Ranker <i>Yeshwantrao Chavan College of Engineering</i> ◦ Secured 1st rank among 214 students by scoring an SGPA of 9.5/10 in 7th Semester. • 3rd Ranker <i>Yeshwantrao Chavan College of Engineering</i> ◦ Secured 3rd rank among 214 students by scoring an SGPA of 9.41/10 in 6th Semester. • Communication Skills Top Performer <i>Yeshwantrao Chavan College of Engineering</i> ◦ Awarded a trophy and certificate for being a top performer in Presentation, Group Discussion, and all aspects of Listening, Speaking, Reading, and Writing skills.	May 2026 Jan 2026 Nov 2025 Apr 2025
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PROFILES

LeetCode

- [Profile](#) 
- Top 12% | Awarded with LeetCode T-Shirt | 30 Badges Earned | 990+ Problems Solved

SKILLS

- **Languages:** C++, Python, SQL
- **Deep Learning & AI:** PyTorch, Hugging Face, HF Transformers and Diffusers, CUDA, timm
- **Cloud Technologies:** AWS
- **DevOps & Version Control:** Git, GitHub
- **Specialized Area:** Deep Learning, Image Processing, Medical AI, Large Language Models, Vision Language Models
- **Mathematical & Statistical Tools:** MATLAB
- **Research Skills:** Literature Review, Academic Writing, $\LaTeX$

LEADERSHIP EXPERIENCE

• Publicity Master <i>Codeware Coding Club</i> ◦ Responsible for leading the Publicity department of the club. ◦ Took decisions on the club's promotion and outreach policies.	Sep 2024 - Nov 2025
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VOLUNTEER EXPERIENCE

• Reviewer <i>IEEE International Conference on Emerging Trends in Engineering and Medical Sciences 2026</i> ◦ Reviewed 8 manuscripts on Artificial Intelligence and Machine Learning.	Nov 2025 - Jan 2026 
• Reviewer <i>IEEE 6th India Council International Conference (INDISCON) 2025, NIT Rourkela</i> ◦ Reviewed 7 manuscripts in the AI & ML, Data Science track.	May 2025 - May 2025 
• Reviewer <i>IEEE 3rd International Symposium on Sustainable Energy, Signal Processing and Cybersecurity 2025</i> ◦ Reviewed 4 manuscripts on Artificial Intelligence and Machine Learning.	Aug 2025 - Aug 2025 

CERTIFICATIONS

• Indian Institute of Science (IISc): KIAC Winter School on AI for FinTech	Jan 2026
• Indian Institute of Science (IISc): EE Summer School	Jul 2025
• Cognitive Class by IBM: Machine Learning with Python	Apr 2025
• Coursera and University of Michigan: Programming for Everybody	Feb 2025
• Cambridge University Press: Cambridge University English Empower Level C1	Feb 2025

PROFESSIONAL MEMBERSHIPS

• IEEE , Membership ID: 100684498	Oct 2024 - Present
• IEEE Computer Society , Membership ID: 100684498	Nov 2025 - Present
• IEEE Signal Processing Society , Membership ID: 100684498	Nov 2025 - Present
• IEEE Communications Society , Membership ID: 100684498	Nov 2025 - Present
• IEEE Women in Engineering , Membership ID: 100684498	Nov 2025 - Present